HR TECHNOLOGY • FULL-STACK WEB APPLICATION

Employee Leave Management System

A secure, role-based platform that streamlines leave requests and approvals for Employees and Managers alike.

React.js | Node.js / Express | PostgreSQL (Supabase) | Vercel + Render

Test Credentials for Evaluation

For reviewers — use these accounts to explore both role-based views

Manager Access

Admin Dashboard

View all employee requests, approve or reject applications, and review detailed employee profiles.

USERNAME / EMAIL

`manager@gcu.in`

PASSWORD

`ZollidMngr#Leave99`

Employee Access

User Portal

Log in as a standard employee to apply for leave and view individual balances. Any account below works.

STANDARD PASSWORD (ALL EMPLOYEES)

`Welcome@2026`

USERNAMES

`michael` `sarah` `david` `emily` `james`

For evaluation purposes only — please rotate or revoke these credentials before any production use.

Project Overview

The Employee Leave Management System is a modern, secure, and fully responsive web application designed to streamline the process of requesting and managing employee time off.

Built with a dual-role architecture, it provides dedicated interfaces for both Employees and Managers — ensuring a frictionless workflow for leave applications, tracking, and approvals.

2	4	100%
USER ROLES	CORE TABLES	RESPONSIVE

Employee Portal

View leave balances, submit requests, and track application history in real time.

Manager Dashboard

Review all requests, approve or reject applications, and explore employee profiles.

Technology Stack

The tools and platforms powering the application

Frontend

React.js with standard CSS — glassmorphism styling and CSS Grid layouts

Backend

Node.js with Express.js powering the REST API and business logic

Database

PostgreSQL, hosted on Supabase for managed, scalable storage

Hosting & Deployment

Vercel for the frontend; Render for the backend service

Role-Based Access Control

Feature 01 — Dedicated experiences for every user type

RBAC

Two tailored interfaces, one platform

Employee Portal

Employees can view their active leave balances, submit leave requests with dynamic reasons, review their request history, and receive real-time notifications on the status of their applications.

Manager Dashboard

Managers get a comprehensive overview of all employee leave requests — approving or rejecting applications, viewing detailed employee profiles, and filtering or sorting requests dynamically.

Feature 02

Advanced Security Measures

Session-Isolated Authentication

SessionStorage replaces standard localStorage, so duplicating or opening a new tab strictly forces re-authentication — preventing unauthorized access on shared devices.

Secure Document Handling

FAQ and Help sections never display sensitive manager credentials. Standard procedures instead route employees to a securely authorized internal document.

Dynamic Leave Application & Notifications

Features 03–04 — Contextual inputs and real-time feedback

Dynamic & Intelligent Application

- **Contextual Dropdowns**

  Selecting a leave type (Sick, Annual) dynamically renders context-appropriate reasons.

- **Custom Inputs**

  An “Other” option spawns a manual text field, giving employees flexibility for specific context.

Real-Time Status Notifications

An in-app notification engine dispatches a color-coded toast the moment a manager acts on a request — delivered on the employee's next login or active session.

✓ **Leave Approved**

✕ **Leave Rejected**

Premium UI/UX & Edge-Case Handling

Feature 05 — Polish that removes friction, not just decoration

Cold Start Mitigation

Free-tier hosting sleeps after inactivity, so the frontend masks the ~60-second wake-up with an animated “Connecting to Server…” state and informational alert.

Centered Profile Modals

Manager interactions use center-aligned modal cards with frosted-glass backgrounds and vibrant theme borders — replacing default browser alerts.

Responsive Grids

Data cards and tables use symmetric CSS grid layouts with max-width limits, looking sharp from mobile phones to ultra-wide monitors.

Database Architecture

PostgreSQL on Supabase — four interconnected tables

Users

Secure credentials, roles (employee / manager), and core profile data.

Leave_Balances

Allotted and remaining days for Annual and Sick leave per employee.

Leave_Requests

Every application: date range, duration, reason, and status (Pending / Approved / Rejected).

Notifications

A ledger of system alerts mapped to specific user_id values for reliable delivery.

All tables are relationally linked through user_id, ensuring balances, requests, and notifications stay consistently in sync for every employee.

Conclusion

This system modernizes an essential HR function by prioritizing both security and user experience.

By implementing strict session controls, graceful loading states, and a beautiful, intuitive interface, it completely removes the friction traditionally associated with leave management.

Security-First

Premium UX

Role-Based Workflow

Real-Time Feedback